LOCAL INSTALLATION PLAYBOOK / WINDOWS 10 & 11

Run AGRO AI on another laptop

A practical guide to installing the project, downloading its local AI models, starting the app, and checking that the API is healthy.

01 INSTALL	02 PREPARE	03 START
Python, Node.js, and Ollama	Extract ZIP and run setup_windows.bat	Download Qwen models and launch

Before you begin

COMPUTER	NETWORK	COST
64-bit Windows 10/11. 16 GB RAM and 10 GB free disk space are recommended for local image and chat models.	Internet access is needed once for installers, Python/npm packages, and multi-GB model downloads.	The default app uses local models. No paid AI API key is required.

The ZIP contains the application source, bundled project model assets, data/catalogue files, setup scripts, and this guide. The larger Ollama language-model weights are downloaded separately on the new laptop.

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1. Requirements and downloads

Install these three programs on the new laptop before running the project setup.

REQUIREMENT	WHAT TO INSTALL / WHY
Ollama for Windows	Install from ollama.com/download/windows. It runs the image and chat models locally.
Python 3.12+	Install from python.org/downloads/windows. Enable PATH integration if offered.
Node.js LTS + npm	Install from nodejs.org. npm is included and installs the React/Vite frontend packages.
Internet + free disk	Required for setup and model downloads. Keep at least 10 GB free for dependencies, caches, and Qwen model weights.

What the setup script installs

Python backend packages come from **backend/requirements.txt**: FastAPI/Uvicorn, Pydantic, SQLAlchemy, HTTPX, Pillow, NumPy, scikit-learn, ONNX Runtime, authentication/security packages, email validation, and python-dotenv. The frontend uses **frontend/package-lock.json** and npm ci to install React, TypeScript, Vite, Tailwind, Axios, and related packages.

Local AI models

The app expects these two Ollama models. Their weights are not embedded in the ZIP; the download script fetches them once on the new laptop.

```
ollama pull qwen2.5vl:3b
ollama pull qwen2.5:1.5b
```

The ZIP also includes the smaller application model files in backend/models and the project datasets. Local .env secrets, databases, upload folders, logs, virtual environments, node_modules, and build caches are excluded.

Default local ports

5173	Web app (Vite)	8000	FastAPI backend	11434	Ollama service

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2. Extract and prepare the project

1. Extract the ZIP to a writable folder. Example: **C:\AGRO_AI**. Do not run the project from inside the ZIP.

2. Open Command Prompt in the extracted folder and run the setup script:

```
cd /d C:\AGRO_AI
setup_windows.bat
```

The script creates a Python virtual environment, installs backend requirements, installs frontend packages with npm ci, copies .env.example to a private .env with a unique local JWT secret, and creates log/upload folders. Initial setup needs internet access and may take several minutes.

Manual dependency commands (only if you prefer to install by hand)

```
py -3.12 -m venv .venv
.venv\Scripts\python.exe -m pip install --upgrade pip setuptools wheel
.venv\Scripts\python.exe -m pip install -r backend\requirements.txt
cd frontend
npm ci
cd ..
copy .env.example .env
```

If using the manual path, edit .env and replace JWT_SECRET_KEY with a unique private value. Never send .env to someone else or put it in the ZIP.

3. Download the local models

Start the Ollama desktop app, then double-click download_models.bat from the project folder. Or run the model commands below in Command Prompt. The first model is multi-GB; allow time for both downloads to finish.

```
ollama --version
ollama list
download_models.bat
```

The model script checks that the local Ollama service is available, pulls qwen2.5vl:3b for image analysis and qwen2.5:1.5b for chat, then prints the installed model list.

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3. Start and verify AGRO AI

With setup and model downloads complete, run the Windows launcher from Command Prompt:

```
cd /d C:\AGRO_AI
start_agro_ai.bat
```

The launcher opens separate backend and frontend windows and opens the web app. Keep both service windows open while using the app. Avoid launching a second copy if the ports are already in use.

Check the app and API

WEB APP	http://localhost:5173
API HEALTH	http://127.0.0.1:8000/api/health
INTERACTIVE API DOCS	http://127.0.0.1:8000/docs
MODEL STATUS	http://127.0.0.1:8000/api/ollama/status

Command-line checks

```
ollama list
curl.exe http://127.0.0.1:8000/api/health
curl.exe http://127.0.0.1:8000/api/ollama/status
```

Expected: API health reports healthy; model status reports the vision and chat models ready; the web app opens at localhost:5173.

First use

Create an account or sign in, then open Crop Analysis or Soil Analysis. Start with a clear, well-lit photo where the leaf, fruit, or soil surface fills most of the frame. First inference may take longer while a model loads into memory.

To stop the web and API services, press Ctrl+C in their terminal windows. The default setup uses local inference and does not require a paid AI API key.

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4. Troubleshooting and handoff

Use these checks in order; keep the backend and frontend windows visible so you can read error messages.

Ollama command is missing

Install Ollama for Windows from the official installer, finish installation, then open a new Command Prompt and run ollama --version.

A model is missing / image or chat returns setup required

Start Ollama and rerun download_models.bat. Confirm both qwen2.5vl:3b and qwen2.5:1.5b appear in ollama list.

API or frontend port is busy

Check whether AGRO AI is already running in another terminal. Stop that copy with Ctrl+C before restarting. Default ports are API 8000 and web 5173. Do not terminate unrelated processes.

Model runner stops / memory error

Close memory-heavy applications and retry with a smaller, clear image. Image analysis runs locally, so speed depends on the laptop CPU/GPU and available RAM. 16 GB RAM is recommended.

Frontend cannot reach the API

Open the API health URL and /docs from the verification table. If health is unavailable, read the backend terminal. If Vite is unavailable, read the frontend terminal.

Useful diagnostic command

```
diagnose_agro_ai.bat
```

Sharing safely

Share the ZIP and this PDF. Keep your local .env, account database, logs, and uploaded photos private. The recipient runs setup_windows.bat, download_models.bat, then start_agro_ai.bat on their own laptop.

QUICK START

```
Install Python 3.12+, Node.js LTS, and Ollama
Extract ZIP to C:\AGRO_AI
setup_windows.bat
download_models.bat
start_agro_ai.bat
Open http://localhost:5173
```

Optional Codespaces and Linux scripts are included in the project. The commands in this guide target the supplied Windows launchers.

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